

Cristian Angel Choque Nacho

Geospatial Data Specialist (GIS | UAV Mapping | Remote Sensing)

GIS: Geodatabases, QA/QC, cartography, spatial analysis for monitoring pipelines.

UAV: Flight planning + photogrammetry (Metashape/Pix4D); orthomosaics, DSM/DTM, point clouds.

Remote Sensing: GEE/ENVI mapping & change detection; Python/ArcPy automation.



Bolivian



Shenzhen, China



cristiangeocon@gmail.com



+86 159 1118 2644



+591 64333870



cacn369



Education

Sep 2024 –
Present

- **Ph.D. Student in Civil Engineering (GIS & Remote Sensing application)**
Shenzhen University, China.
Research focus: Geospatial hazard monitoring, multi-source GIS & Remote Sensing.

Sep 2022 –
Jul 2024

- **M.Sc. in Remote Sensing & GIS (Space Technology Applications)**
Beihang University, Beijing, China.

2014 –
2019

- **B.Sc. in Geographical Engineering**
Military School of Engineering, La Paz, Bolivia.
Graduated with Honors (Approval with Excellence).

Professional Experience

Jan 2020 –
Jan 2023

- **Geospatial Data Specialist (Contract).** Municipal Government of La Paz City (GAMLP), Bolivia.
Unit: Municipal Regulation and Supervision System (SIREMU).

- Owned and updated an operational **GIS geodatabase** for urban-waste supervision, integrating routes, service areas, and facility/asset inventories with concessionary operators.
- Performed **spatial analysis** to support supervision planning, verify service coverage, and inform route and operation adjustments.
- Planned and executed **UAV mapping** campaigns and produced survey-grade deliverables (orthomosaics, DSM/DTM, point clouds, contours) to update municipal cartography.
- Produced technical reports, thematic maps, and landfill **volumetric calculations** to support engineering and operational decisions.

Jan 2022 –
March 2022

- **Geospatial Data Analyst (Contract).** Amazon Regional Observatory (ORA) – Boliviana de Innovación Tecnológica Satelital (BITS), Bolivia.
Project: Forest Deforestation Platform, Amazon Regional Observatory (ORA).

- Processed and standardized **forest/deforestation** geospatial indicators for publication and web visualization on the ORA platform.
- Managed dataset ingestion to server-side **geodatabases**, ensuring consistent schemas, metadata, and multi-temporal organization.
- Supported environmental monitoring by integrating multi-source geographic layers and validating data quality prior to release.

Professional Experience (continued)

Feb 2019 –
Nov 2019

- **UAV photogrammetry mapping Analyst (Internship).** ASTRO Geomatics, Bolivia.
Project: Photogrammetric Research & Implementation.
 - Planned and executed rotary-wing UAV surveys for topographic mapping across university campuses and adjacent sites.
 - Established and densified **GCP networks** using static GNSS to ensure accurate georeferencing and quality control.
 - Processed imagery in **Agisoft Metashape** and **Pix4D** to deliver orthomosaics, DSM/DTM, point clouds, and 3D models.

Technical Skills

GIS &
Geospatial

- **Advanced:** ArcGIS Pro/Desktop, QGIS, Google Earth Engine (GEE).
Intermediate: AutoCAD, Global Mapper.
Core: Spatial analysis, geodatabase management, coordinate systems, cartography, QA/QC.

UAV Mapping &
Photogramme-
try

- **Software:** Agisoft Metashape, Pix4D Mapper.
Field: Flight planning & execution (rotary-wing UAV), GNSS/GPS (RTK & static), Total Station, GCP workflows.
Outputs: Orthomosaics, Point Clouds, DSM/DTM, Contours, 3D Modeling.

Remote Sensing
(Satellite)

- **Software/Platforms:** Python, R, Google Earth Engine (GEE), ENVI.
Tasks: Image preprocessing, feature extraction, land-cover mapping, change detection.

Programming &
Automation

- **Python:** ArcPy, Pandas; geospatial scripting and workflow automation.
R: spatial analysis (RStudio).

Languages

- **Spanish** (Native), **English** (C1), **Chinese** (HSK 3), **Japanese** (N5).

Certifications & Training

Technical

2026

- **Claude Code in Action.** Anthropic Education
- **Claude Code 101.** Anthropic Education

2025

- **Artificial Intelligence for Civil Engineers.** Ph.D. coursework, Shenzhen University.

2024

- **Geographic Information System: Design and Practice.** M.Sc., Beihang University.
- **Deep Learning for Remote Sensing Image Processing.** M.Sc., Beihang University.
- **Natural Disaster Remote Sensing.** M.Sc., Beihang University.

2023

- **Spatial Analysis & Google Earth Engine with R, MASTERGIS.**
- **Python Programming in ArcGIS, MASTERGIS.**
- **Environmental Impact Assessment with ArcGIS, MASTERGIS.**
- **Generative AI Overview for Project Managers, PMI.**

2021

- **SfM Photogrammetry with Metashape, ASTRO Geomatics.**

Conferences

2023

- **6th International Workshop on GIS Technology**, Beijing.

2019

- **3rd Aerospace Bolivian Conference**, IEEE.

2014–present

- *Ongoing professional development through +40 additional courses and workshops.*